Laboratory Course: [BCS-351] Data Structures Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: VIVA QUESTIONS**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Dr. R. K. Sharma (Associate**

Viva Voce Top 50 Questions & Model Answers: Data Structures Lab

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1. Stacks, Queues & Recursion Viva Questions

- Q1: What is the main disadvantage of a linear queue implemented via an array?

  A: False overflow condition occurs when elements are dequeued from front, leaving unused space.

- Q2: How does a circular queue overcome false overflow?

  A: By wrapping rear and front pointers using modulo arithmetic: rear = (rear + 1) % MAX.

- Q3: How does the system call stack facilitate recursion?

  A: Each recursive invocation allocates an activation record containing local variables and return address.

2. Trees, Graphs & Complexity Viva Questions

- Q4: What is the worst-case search complexity in an unbalanced BST vs an AVL tree?

  A: Unbalanced BST is O(n) (skewed tree); AVL tree maintains balance factor {-1,0,1} guaranteeing O(log n).

- Q5: Why is Quick Sort preferred over Merge Sort for arrays, and vice-versa for linked lists?

  A: Quick Sort has superior cache locality and in-place partitioning; Merge Sort does not require random access.